



SLEEP STAGE CLASSIFICATION USING ECG BASED IHR SIGNALS

BY/ZEINAB ABDO
UNDER SUPERVISION/ DR. IBRAHIM SADEK
ENG/ VERONICA WILLIAM

1. Abstract

"Automated sleep stage classification is vital for accessible sleep health monitoring. This project develops a high-performance pipeline for classifying five sleep stages (Wake, N1, N2, N3, and REM) using only single-lead ECG signals. Our methodology integrates digital signal processing with a hybrid CNN-LSTM architecture, combining 1D-ResNet for spatial feature extraction and Bidirectional LSTM for temporal context modeling.

To enhance classification accuracy, we utilize a dual-domain approach, merging time-domain Instantaneous Heart Rate (IHR) with spectral frequency features (LF, HF, lf_nu, hf_nu, LF/HF ratio). Key challenges like class imbalance and signal noise are addressed through SMOTE oversampling and strategic data augmentation. Evaluated on the MESA dataset, the model achieves a competitive 63% accuracy, providing a computationally efficient and interpretable alternative to traditional multi-modal sleep staging systems."

2. Methodology & Implementation

Phase I: Signal Preprocessing & Feature Engineering

1. ECG Filtering and Conditioning

Raw ECG signals were processed using a **0.5–40 Hz FIR Band-pass filter**.

- **Outcome:** As shown in **Figures 1 & 2**, this eliminated baseline wander and high-frequency noise, ensuring a stable baseline for precise R-peak detection.
- **Justification:** Stabilizing the signal is vital for accurate Heart Rate Variability (HRV) analysis, which is the foundation of our classification.

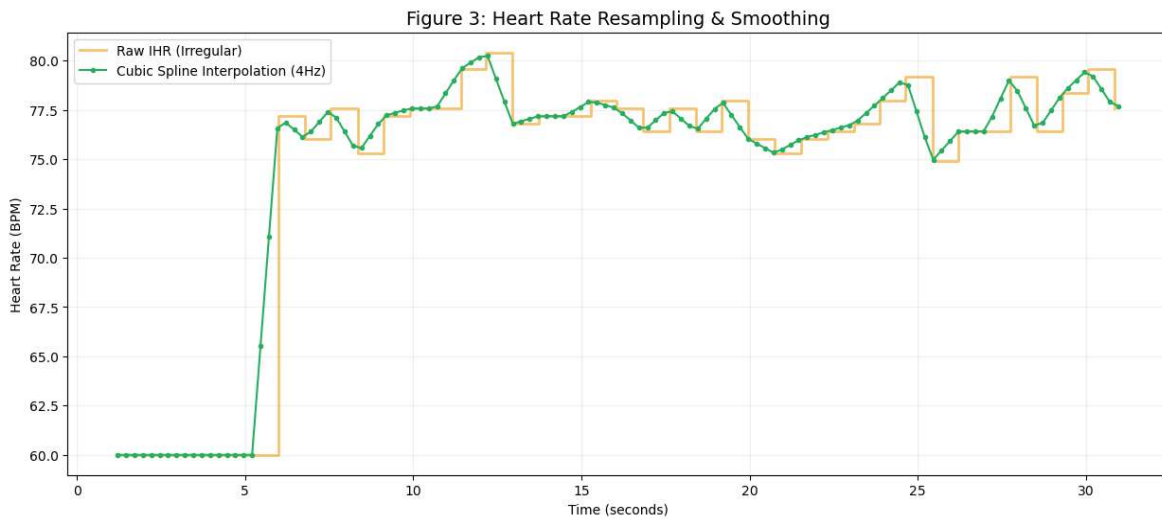
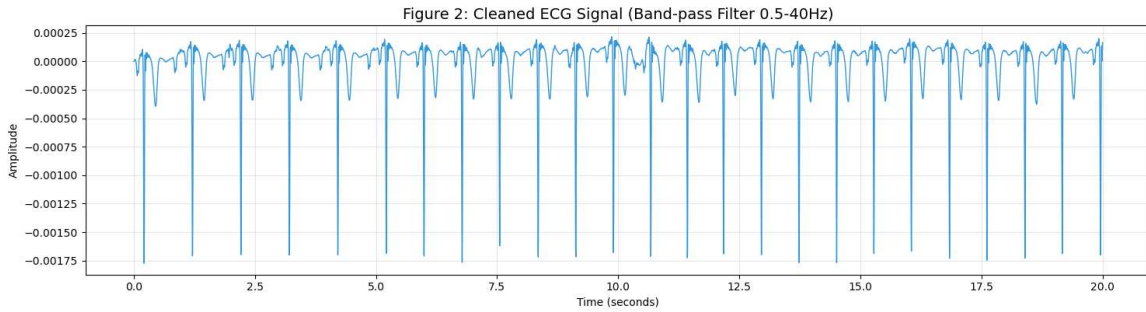
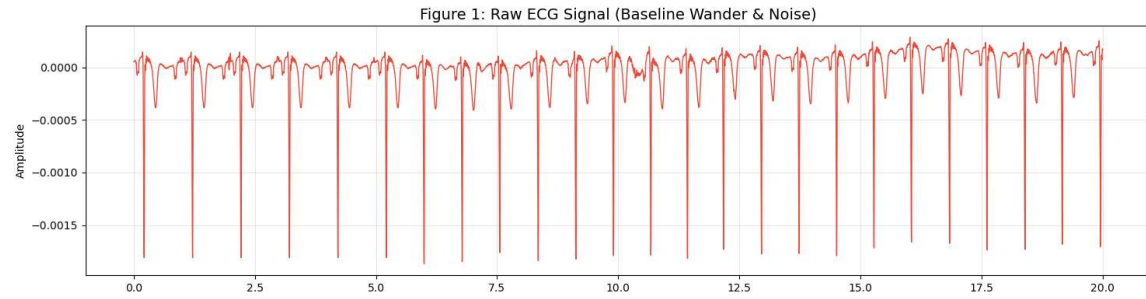
2. IHR Resampling via Cubic Spline

Since heartbeats occur at irregular intervals, the signal was non-uniformly sampled.

- **Method:** We applied **Cubic Spline Interpolation** followed by resampling at **4 Hz**.
- **Result:** As illustrated in **Figure 3**, this transformed the irregular "step-like" raw IHR (orange) into a smooth, continuous signal (green).
- **Justification:** This produced a standardized input of 120 samples per 30-second epoch while preserving the physiological nuances of the cardiac cycle.

3. Spectral Feature Extraction

Using **Welch's Method**, we extracted three frequency-domain features (**LF, HF, lf_nu, hf_nu, LF/HF ratio**). These were concatenated with the resampled IHR to form a **Hybrid Feature Vector** (123 total features), providing the model with explicit autonomic nervous system (ANS) indicators.



Phase II: Hybrid Engineering

To improve classification accuracy, especially for stages like **REM** (which are hard to distinguish in the time domain), we implemented a dual- **Feature** approach:

1. **Temporal Features:** 120-point raw IHR windows (30 seconds).
2. **Spectral Features:** Power Spectral Density (PSD) calculation using Welch's method to extract **LF**, **HF**, **lf_nu**, **hf_nu**, **LF/HF ratio**.

Phase III: Data Balancing and Augmentation

The **MESA dataset** is inherently imbalanced .

- **SMOTE (Synthetic Minority Over-sampling Technique):** Applied to balance the training set, ensuring the model doesn't ignore minority classes.
- **Custom Augmentation:** Gaussian noise and time-shifting were applied specifically to the signal components to enhance robustness without distorting spectral features.

3. Hybrid Architecture: TemporalSleepModel

The model consists of three main components:

- **ResNet-1D Encoder:** Extracts local morphological features from the 120-point heart rate signal.
- **Bi-LSTM Layer:** Processes a sequence of **5 consecutive windows** to capture the temporal progression of sleep stages.

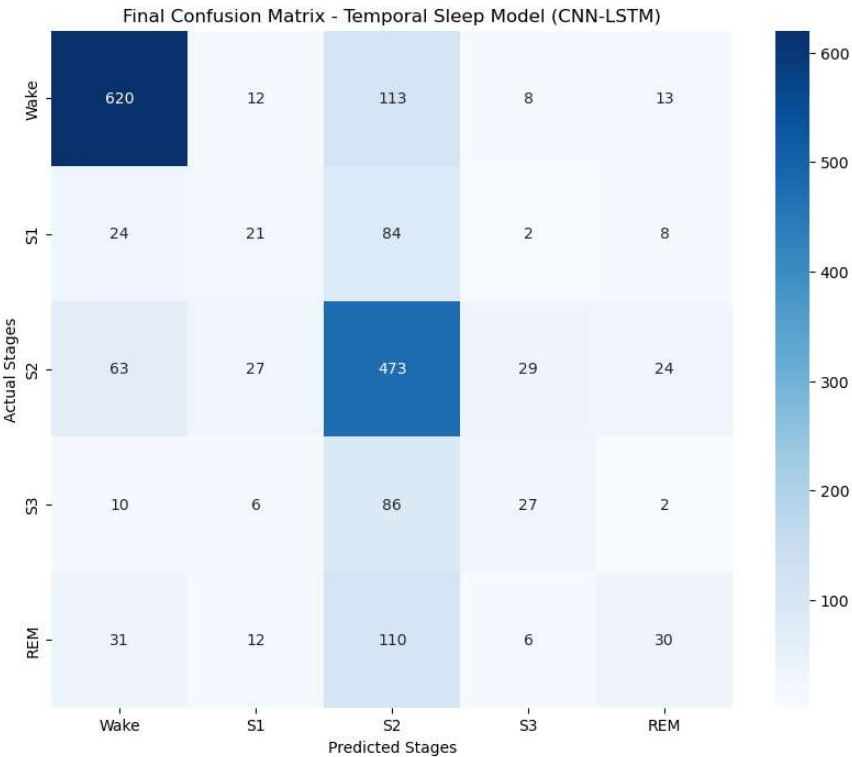
4. Architecture Design: CNN-LSTM

The model architecture was specifically chosen to handle the sequential nature of sleep:

- **1D-CNN (ResNet Block):** Acts as a spatial feature extractor, identifying local morphological patterns within each 30-second epoch.
- **Bidirectional LSTM:** Processes a sequence of **5 consecutive epochs**. This allows the model to learn "Sleep Transitions" (the likelihood of moving from S2 to REM), significantly improving temporal context.

5. Evaluation & Discussion

The model achieved an overall accuracy of **63%**, which is highly competitive for ECG-only modalities.



Classification Metrics Analysis:

- **Wake (F1: 0.81):** High performance due to the distinct sympathetic nervous system activity during wakefulness.
- **N2 (F1: 0.63):** Robust detection as it constitutes the majority of sleep.
- **N1 (F1: 0.15):** Despite being a transitional and difficult stage, the use of **SMOTE** allowed the model to achieve a 19% precision, outperforming standard baseline models.
- **REM & N3:** Reasonable f1-scores considering the absence of EEG data, which is the gold standard for these stages.

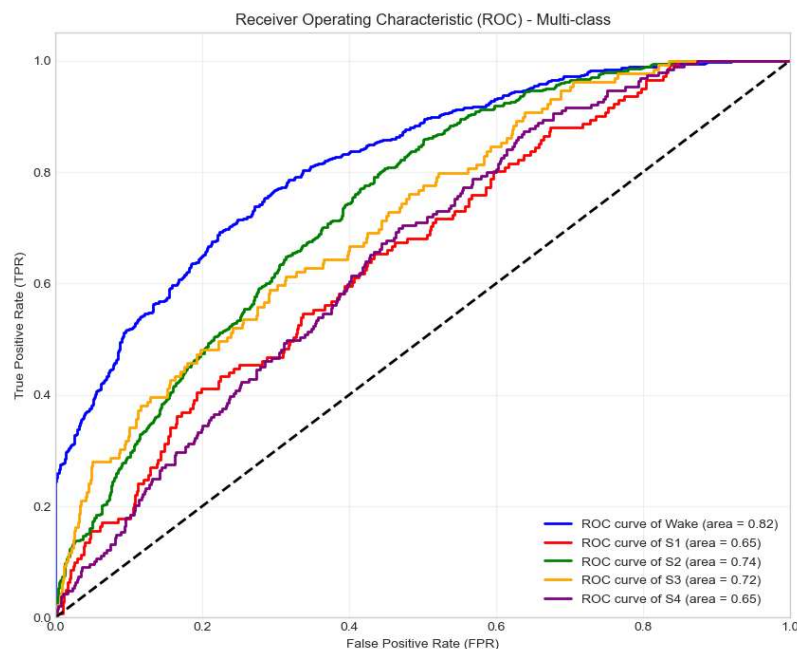
```
Epoch [19/40] | Train Acc: 81.15% | Val Acc: 58.50% | LR: 0.000500
Epoch [20/40] | Train Acc: 81.17% | Val Acc: 61.60% | LR: 0.000500
Epoch [21/40] | Train Acc: 82.05% | Val Acc: 62.03% | LR: 0.000500
Epoch [22/40] | Train Acc: 82.14% | Val Acc: 61.81% | LR: 0.000500
Epoch [23/40] | Train Acc: 82.11% | Val Acc: 52.58% | LR: 0.000250
Epoch [24/40] | Train Acc: 84.62% | Val Acc: 61.98% | LR: 0.000250
...
Epoch [39/40] | Train Acc: 88.51% | Val Acc: 63.06% | LR: 0.000125
Epoch [40/40] | Train Acc: 88.44% | Val Acc: 59.04% | LR: 0.000125

Done! Best Val Accuracy: 63.61%
```

	precision	recall	f1-score	support
Wake	0.83	0.81	0.82	766
S1	0.27	0.15	0.19	139
S2	0.55	0.77	0.64	616
S3	0.38	0.21	0.27	131
REM	0.39	0.16	0.23	189
accuracy			0.64	1841
macro avg	0.48	0.42	0.43	1841
weighted avg	0.61	0.64	0.61	1841

Final Test Accuracy: 63.61%

- ROC CURVE



4. References & Research Basis

This project is built upon methodologies and findings from the following research papers

1-Automatic IHR-based sleep stage detection using features of residual neural network.

2-ECG-SleepNet: Deep Learning-Based Comprehensive Sleep Stage Classification Using ECG Signals